

Отчёт по лабораторной работе 4

Первоначальное конфигурирование сети

Седохин Даниил Алексеевич

Содержание

1	Цель работы	5
2	Задание	6
3	Контрольные вопросы	12
4	Выводы	13

Список иллюстраций

2.1	Размещение коммутаторов и оконечных устройств согласно схеме .	6
2.2	Настройка коммутатора msk-donskaya-dasedokhin-sw-1	7
2.3	Настройка коммутатора msk-donskaya-dasedokhin-sw-2	8
2.4	Настройка коммутатора msk-donskaya-dasedokhin-sw-3	9
2.5	Настройка коммутатора msk-donskaya-dasedokhin-sw-4	10
2.6	Настройка коммутатора msk-pavlovskaya-dasedokhin-sw-1	11

Список таблиц

1 Цель работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети.

2 Задание

В логической рабочей области Packet Tracer разместите коммутаторы и оконечные устройства согласно схеме сети L1 (см. рис. 3.1 из раздела 3.3) и соедините их через соответствующие интерфейсы (рис. 2.1).

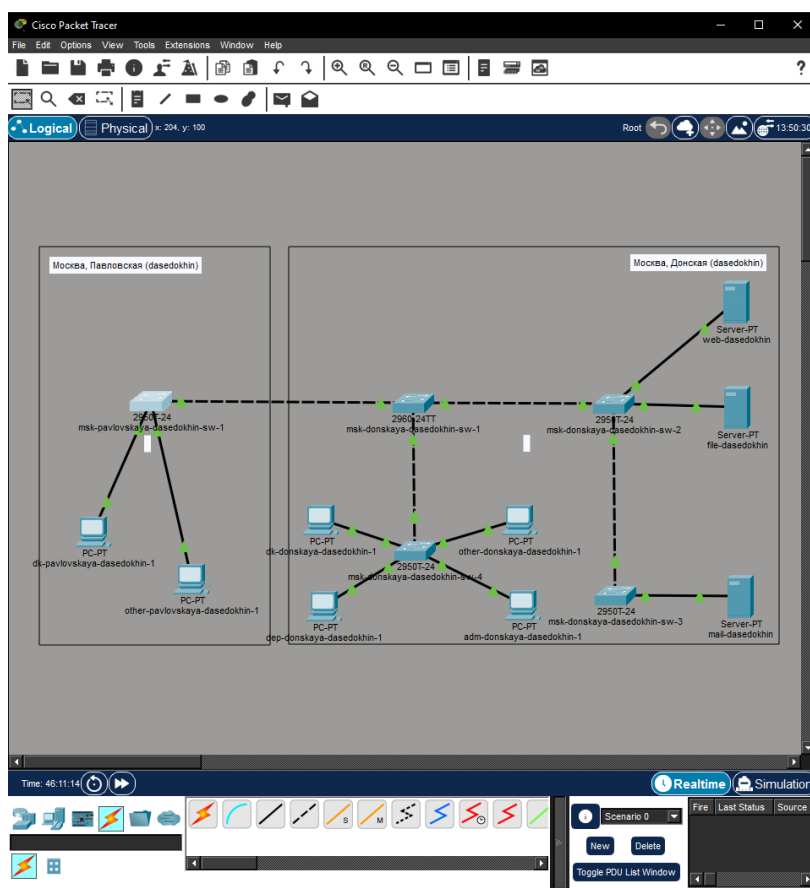
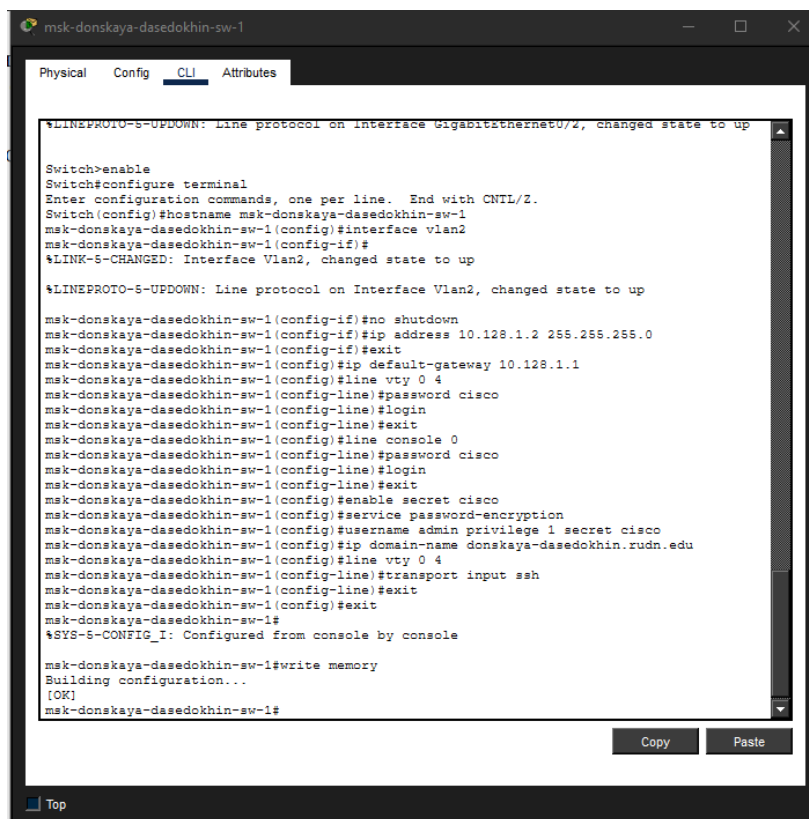


Рисунок 2.1: Размещение коммутаторов и оконечных устройств согласно схеме

Теперь нам необходимо провести первоначальную настройку коммутаторов

нашей сети.

Настроим каждый коммутатор: (рис. 2.2).



```
msk-donskaya-dasedokhin-sw-1
Physical Config CLI Attributes
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dasedokhin-sw-1
msk-donskaya-dasedokhin-sw-1(config)#interface vlan2
msk-donskaya-dasedokhin-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-dasedokhin-sw-1(config-if)#no shutdown
msk-donskaya-dasedokhin-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dasedokhin-sw-1(config-if)#exit
msk-donskaya-dasedokhin-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-1(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-1(config-line)#password cisco
msk-donskaya-dasedokhin-sw-1(config-line)#login
msk-donskaya-dasedokhin-sw-1(config-line)#exit
msk-donskaya-dasedokhin-sw-1(config)#line console 0
msk-donskaya-dasedokhin-sw-1(config-line)#password cisco
msk-donskaya-dasedokhin-sw-1(config-line)#login
msk-donskaya-dasedokhin-sw-1(config-line)#exit
msk-donskaya-dasedokhin-sw-1(config)#enable secret cisco
msk-donskaya-dasedokhin-sw-1(config)#service password-encryption
msk-donskaya-dasedokhin-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-dasedokhin-sw-1(config)#ip domain-name donskeya-dasedokhin.rudn.edu
msk-donskaya-dasedokhin-sw-1(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-1(config-line)#transport input ssh
msk-donskaya-dasedokhin-sw-1(config-line)#exit
msk-donskaya-dasedokhin-sw-1(config)#exit
msk-donskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-dasedokhin-sw-1#
```

Рисунок 2.2: Настройка коммутатора msk-donskaya-dasedokhin-sw-1

Настройка коммутатора msk-donskaya-dasedokhin-sw-2 (рис. 2.3).

```
msk-donskaya-dasedokhin-sw-2
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/24, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dasedokhin-sw-2
msk-donskaya-dasedokhin-sw-2(config)#interface vlan2
msk-donskaya-dasedokhin-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dasedokhin-sw-2(config-if)#no shutdown
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-dasedokhin-sw-2(config-if)#exit
msk-donskaya-dasedokhin-sw-2(config)#ip default gateway 10.128.1.1

% Invalid input detected at '^' marker.

msk-donskaya-dasedokhin-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-2(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-2(config-line)#password cisco
msk-donskaya-dasedokhin-sw-2(config-line)#login
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#line console 0
msk-donskaya-dasedokhin-sw-2(config-line)#password cisco
msk-donskaya-dasedokhin-sw-2(config-line)#login
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#enable secret cisco
msk-donskaya-dasedokhin-sw-2(config)#service password-encryption
msk-donskaya-dasedokhin-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-dasedokhin-sw-2(config)#ip domain name donskeya-dasedokhin.rudn.edu
msk-donskaya-dasedokhin-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dasedokhin-sw-2.donskeya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dasedokhin-sw-2(config)#line vty 0 4
*Mar 2 22:20:43.704: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:20:43.704: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dasedokhin-sw-2(config-line)#transport input ssh
msk-donskaya-dasedokhin-sw-2(config-line)#exit
msk-donskaya-dasedokhin-sw-2(config)#exit
msk-donskaya-dasedokhin-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-dasedokhin-sw-2#
```

Рисунок 2.3: Настройка коммутатора msk-donskaya-dasedokhin-sw-2

Настройка коммутатора msk-donskaya-dasedokhin-sw-3 (рис. 2.4).


```
msk-donskaya-dasedokhin-sw-3
Physical Config CLI Attributes
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname dontskaya-dasedokhin-sw-3
dontskaya-dasedokhin-sw-3(config)#interface vlan2
dontskaya-dasedokhin-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

dontskaya-dasedokhin-sw-3(config-if)#no shutdown
dontskaya-dasedokhin-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
dontskaya-dasedokhin-sw-3(config-if)#exit
dontskaya-dasedokhin-sw-3(config)#ip default-gateway 10.128.1.1
dontskaya-dasedokhin-sw-3(config)#line vty 0 4
dontskaya-dasedokhin-sw-3(config-line)#password cisco
dontskaya-dasedokhin-sw-3(config-line)#login
dontskaya-dasedokhin-sw-3(config-line)#exit
dontskaya-dasedokhin-sw-3(config)#line console 0
dontskaya-dasedokhin-sw-3(config-line)#password cisco
dontskaya-dasedokhin-sw-3(config-line)#login
dontskaya-dasedokhin-sw-3(config-line)#exit
dontskaya-dasedokhin-sw-3(config)#enable secret cisco
dontskaya-dasedokhin-sw-3(config)#service password-encryption
dontskaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
^
% Invalid input detected at '^' marker.

dontskaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
^
% Invalid input detected at '^' marker.

dontskaya-dasedokhin-sw-3(config)#username admin privilege 1 secret cisco
dontskaya-dasedokhin-sw-3(config)#ip domain-name dontskaya-dasedokhin.rudn.edu
dontskaya-dasedokhin-sw-3(config)#line vty 0 4
dontskaya-dasedokhin-sw-3(config-line)#exit
dontskaya-dasedokhin-sw-3(config)#crypto key generate rsa
The name for the keys will be: dontskaya-dasedokhin-sw-3.dontskaya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

dontskaya-dasedokhin-sw-3(config)#line vty 0 4
*Mar 2 22:26:35.720: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:26:35.720: %SSH-5-ENABLED: SSH 1.5 has been enabled
dontskaya-dasedokhin-sw-3(config-line)#transport input ssh
dontskaya-dasedokhin-sw-3(config-line)#exit
dontskaya-dasedokhin-sw-3(config)#exit
dontskaya-dasedokhin-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

dontskaya-dasedokhin-sw-3#write memory
Building configuration...
[OK]
dontskaya-dasedokhin-sw-3#
```

Рисунок 2.4: Настройка коммутатора msk-donskaya-dasedokhin-sw-3

Настройка коммутатора msk-donskaya-dasedokhin-sw-4 (рис. 2.5).

```
msk-donskaya-dasedokhin-sw-4
Physical Config CLI Attributes

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#configure terminal
^
% Invalid input detected at '^' marker.

Switch(config)#hostname msk-donskaya-dasedokhin-sw-4
msk-donskaya-dasedokhin-sw-4(config)#interface vlan2
msk-donskaya-dasedokhin-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-dasedokhin-sw-4(config-if)#no shutdown
msk-donskaya-dasedokhin-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-dasedokhin-sw-4(config-if)#exit
msk-donskaya-dasedokhin-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-dasedokhin-sw-4(config)#line vty 0 4
msk-donskaya-dasedokhin-sw-4(config-line)#password cisco
msk-donskaya-dasedokhin-sw-4(config-line)#login
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#line console
% Incomplete command.
msk-donskaya-dasedokhin-sw-4(config)#line console 0
msk-donskaya-dasedokhin-sw-4(config-line)#password cisco
msk-donskaya-dasedokhin-sw-4(config-line)#login
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#enable secret cisco
msk-donskaya-dasedokhin-sw-4(config)#service password-encryption
msk-donskaya-dasedokhin-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-dasedokhin-sw-4(config)#ip domain-name donskeya-dasedokhin.rudn.edu
msk-donskaya-dasedokhin-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dasedokhin-sw-4.donskeya-dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dasedokhin-sw-4(config)#line vty 0 4
*Mar 2 22:30:25.817: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 2 22:30:25.817: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dasedokhin-sw-4(config-line)#transport input ssh
msk-donskaya-dasedokhin-sw-4(config-line)#exit
msk-donskaya-dasedokhin-sw-4(config)#exit
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-dasedokhin-sw-4#
```

Рисунок 2.5: Настройка коммутатора msk-donskaya-dasedokhin-sw-4

Настройка коммутатора msk-pavlovskaya-dasedokhin-sw-1 (рис. 2.6).

```
msk-pavlovskaya-dasedokhin-sw-1
Physical Config CLI Attributes

Switch>enable
Switch>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-pavlovskaya-dasedokhin-sw-1
msk-pavlovskaya-dasedokhin-sw-1(config)#interface vlan2
msk-pavlovskaya-dasedokhin-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-pavlovskaya-dasedokhin-sw-1(config-if)#no shutdown
msk-pavlovskaya-dasedokhin-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-dasedokhin-sw-1(config-if)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-dasedokhin-sw-1(config)#line vty 0 4
msk-pavlovskaya-dasedokhin-sw-1(config-line)#password cisco
msk-pavlovskaya-dasedokhin-sw-1(config-line)#login
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#line console 0
msk-pavlovskaya-dasedokhin-sw-1(config-line)#password cisco
msk-pavlovskaya-dasedokhin-sw-1(config-line)#login
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#enable secret cisco
msk-pavlovskaya-dasedokhin-sw-1(config)#service password-encryption
msk-pavlovskaya-dasedokhin-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-dasedokhin-sw-1(config)#ip domain-name pavlovskaya-dasedokhin.rudn.edu
msk-pavlovskaya-dasedokhin-sw-1(config)#crypto key generate rsa
% You already have RSA keys defined named msk-donskaya-dasedokhin-sw-1.donskaya-
dasedokhin.rudn.edu .
% Do you really want to replace them? [yes/no]: no
msk-pavlovskaya-dasedokhin-sw-1(config)#crypto key generate rsa
% You already have RSA keys defined named msk-donskaya-dasedokhin-sw-1.donskaya-
dasedokhin.rudn.edu .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: msk-pavlovskaya-dasedokhin-sw-1.pavlovskaya-
dasedokhin.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-dasedokhin-sw-1(config)#line vty 0 4
*Mar 1 0:3:33.186: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:3:33.186: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-dasedokhin-sw-1(config-line)#transport input ssh
msk-pavlovskaya-dasedokhin-sw-1(config-line)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#exit
msk-pavlovskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
w
Building configuration...
[OK]
msk-pavlovskaya-dasedokhin-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-dasedokhin-sw-1#
```

Рисунок 2.6: Настройка коммутатора msk-pavlovskaya-dasedokhin-sw-1

3 Контрольные вопросы

1. При помощи каких команд можно посмотреть конфигурацию сетевого оборудования?

`show running-config`

2. При помощи каких команд можно посмотреть стартовый конфигурационный файл оборудования?

`show startup-config`

3. При помощи каких команд можно экспортировать конфигурационный файл оборудования?

`copy running-config startup-config write memory`

`copy running-config tftp:`

`copy running-config flash:имя_файла.txt`

4. При помощи каких команд можно импортировать конфигурационный файл оборудования?

`copy tftp: running-config`

`copy flash:имя_файла.txt running-config`

`copy startup-config running-config`

4 Выводы

Я приобрел навыки по первоначальной неастройке коммутаторов сети.